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ON SCENE COORDINATOR'S REPORT

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CERCLA IMMEDIATE REMOVAL PROJECT

NO. 68-95-007

NORTHERNAIRE PLATING COMPANY

CADILLAC, MICHIGAN

Region V
Environmental Services Division
Spill Response Section



EPA Region 5 Records Ctr.



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KEN BANASZEK, ON-SCENE COORDINATOR

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1.0 SUMMARY OF EVENTS

1.1 Location

The Northernnaire Plating Company is located at 1002 Sixth Street in Cadillac, Wexford County, Michigan (Figure 1). The site is one half mile northwest of the center of the City of Cadillac, in an area of mixed residential, commercial, and industrial facilities. The nearest building is leased by a beer distributor and is 250 feet west of the site. The site is located one fourth of a mile southwest of the City Of Cadillac's municipal well field. The topography of the immediate area is flat and dominated by sandy soils. The property is owned by R.W. Meyer, Inc., Cadillac, Michigan who leased the facility to the Northernnaire Plating Company. Mr. Willard Garwood was President of Northernnaire Plating Company.

1.2 Initial Situation

The Northernnaire Plating Company began operations in 1971. Several private wells near the Northernnaire Plating Company were tested by the Michigan Department of Natural Resources (MDNR) in October 1978. High levels of chromium were found in two wells. The source of soil and ground water contamination was identified as being a leaking sewer line and catch basin north of the Northernnaire Plating Company building. Plating waste was discharged into the sewer line from the facility's process tanks. Sewer line connections were improperly constructed and waste seeped into the ground at several locations. In October 1978, the City of Cadillac revoked Northernnaire's sewer discharge permit and plugged their sewer connection. It is suspected that plating waste continued to be discharged into the sewer line and seeped into the ground. Northernnaire continued operations until April or May 1981.

The MDNR inspected the site in May 1981 and found the facility deserted. Drums of plating waste were left outside the building. A child playing near the barrels sustained a chemical burn from the material contained in one of the drums. MDNR personnel transferred the barrels inside the building.

On 16 and 17 March, 1983 a site inspection was conducted by the Technical Assistance Team (TAT), U.S. EPA, and MDNR. The inspection was made to confirm the MDNR report indicating possible gas vapors inside the building.

Air monitoring was conducted by U.S. EPA and TAT personnel inside and outside the building using CN Draeger tubes, an

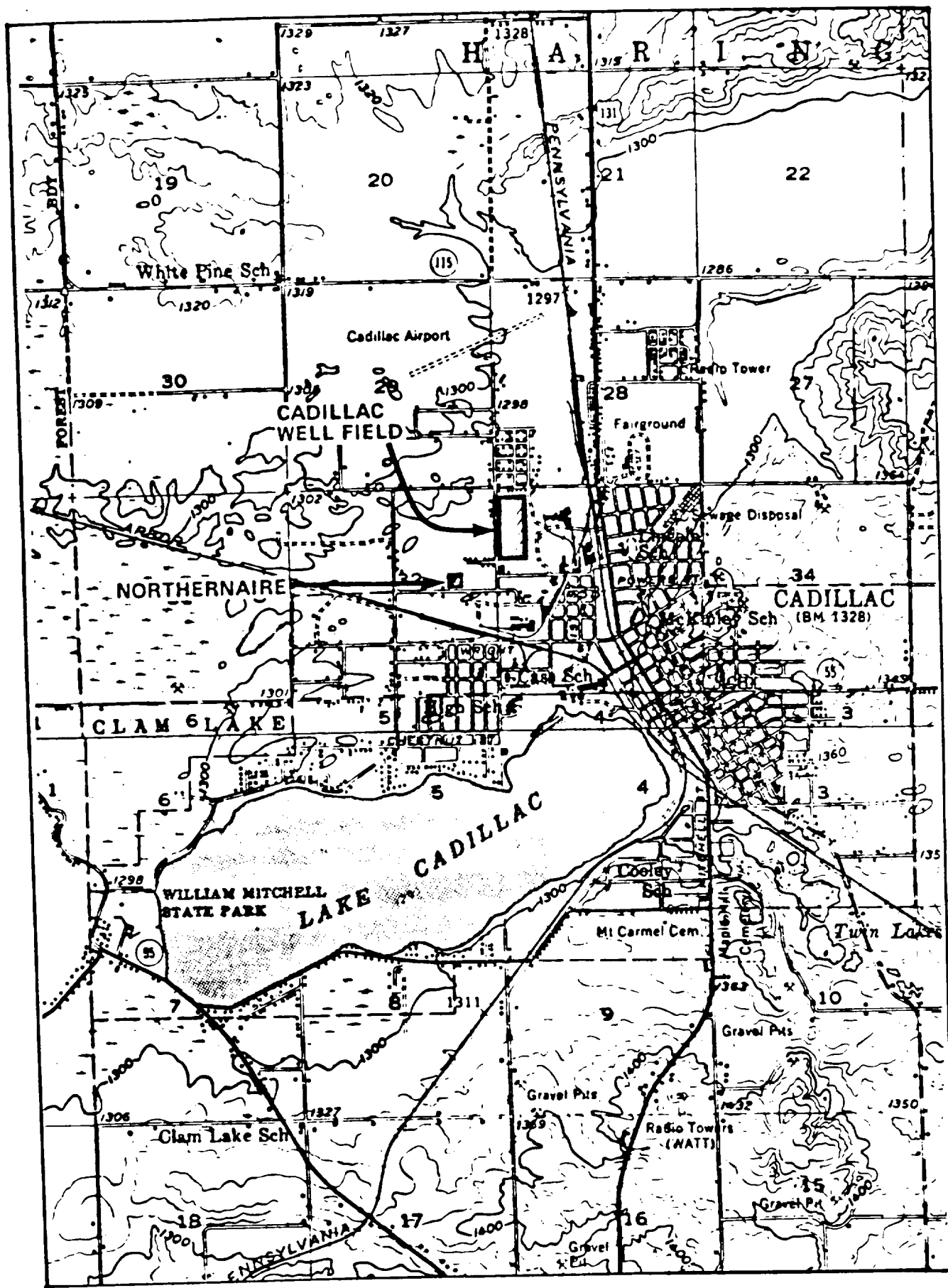


Figure 1: Cadillac area and Northernnaire location.

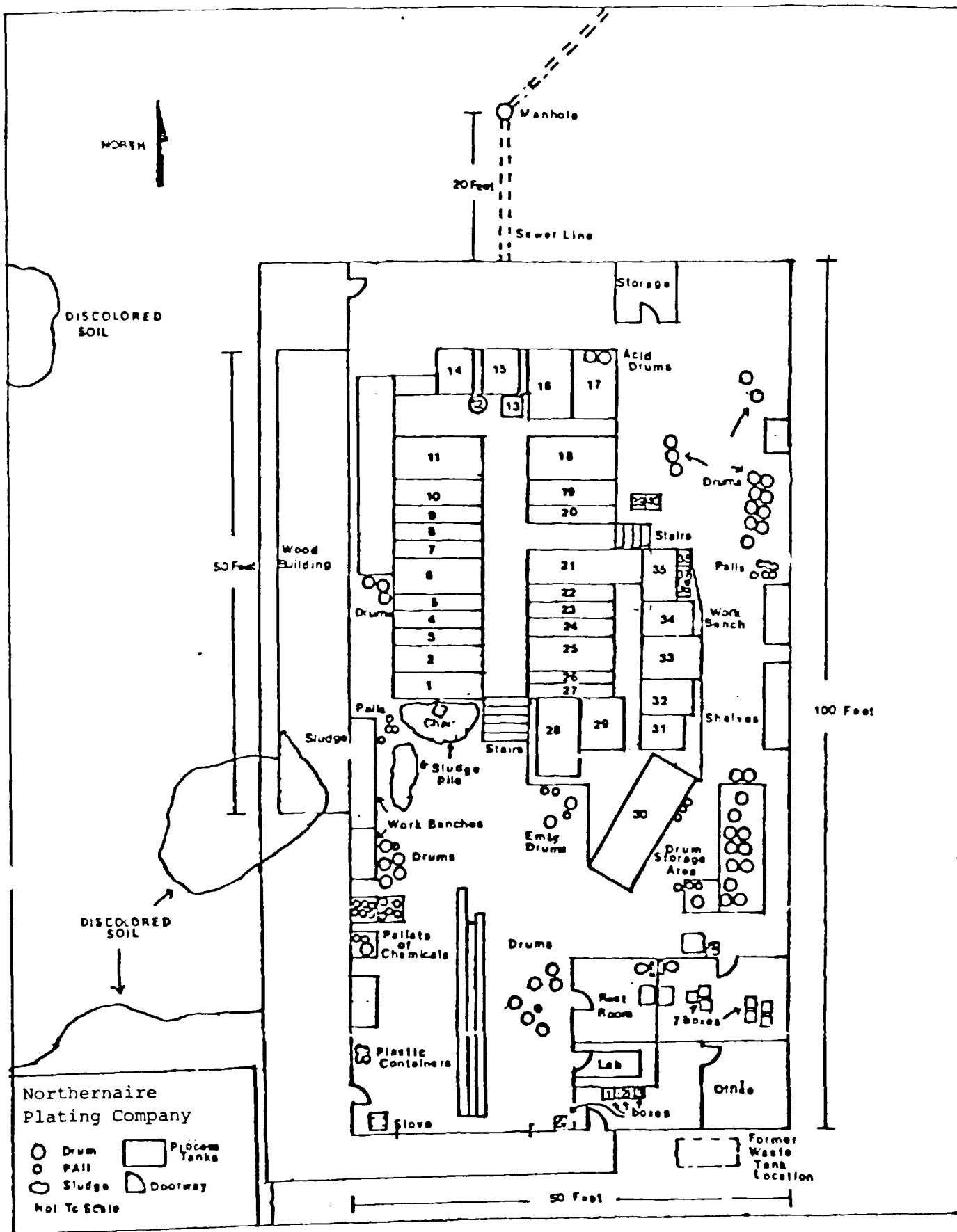


Figure 2. Northernnaire Plating Co. building detail. Numbers correspond to site inventory contained in Appendix Q.

explosimeter and HNU. No readings were obtained inside or outside the building. A site survey was made in Level "8" protection by TAT inspectors. The inside of the building was in a state of disarray, however, no leaking drums or tanks were observed. Liquids were observed in several process tanks inside the building. Drum, tank and lab chemical locations are identified in Figure 2. A completed inventory of tanks and pH measurements are listed in Appendix Q. An inventory of cardboard boxes is also contained in Appendix Q.

TAT inspectors located a sewer line on the north side of the building. The inspectors opened the catch basin twenty feet north of the building and observed two pipes and discolored soil. The sewer line was not sealed and the bottom of the catch basin opened to the ground. It was apparent that waste was being discharged into the catch basin where it seeped into the ground. As the ground became saturated, the material entered the second pipe that drained northeast into a sanitary sewer line. The sanitary sewer line flowed along Eighth Street and discharged to the City of Cadillac Sewage Treatment Plant.

The Federal OSC determined that the site was not an immediate threat to human health or the environment but recommended that the liquids, drums and sludge be removed as soon as possible to prevent future discharges from the facility. (SEE TAT TDD# 5-8303-5, entitled "Site Inspection for Northern Plating Company - Cadillac, Michigan April 1983").

1.3 Federal Cleanup Action

A federal Immediate Removal Action was initiated on 5 July 1983 when a site visit was made and samples were collected by Petrochem Services and their subcontractor Environmental Consultants, Inc. All process tanks, drums and small containers were numbered and samples were collected for waste characterization. The waste material would be bulked for disposal based upon compatibility. The sampling was concluded on 8 July 1983, when the samples were delivered to ECI's laboratory for analysis. Appendix Q contains a summary of the sampling results. Additional work completed during this week included installation of utilities, office trailer, and construction of a concrete decontamination pad.

During the second week, commencing 11 July 1983, Petrochem began the cleanup of contaminated material from the building. The material was decontaminated on the concrete pad using a hypochlorite rinse. The rinse water was collected on the pad and pumped into a vac truck for disposal at Nelson Industrial Services, Inc. of Detroit. The decontamination of the Class

III refuse continued through the week. Additional work included: securing Northernnaire office records for the U.S. EPA Enforcement Division; arrangement for state and RCRA-approved disposal facilities; and procurement of neutralization material for the solidification of floor and tank sludge.

The third week of the cleanup began with the removal of acid wastes from process tanks and containers. The acid liquid was bulked and also transferred to Nelson Industrial Services, a RCRA-approved disposal facility. The acid sludge was removed from each tank, neutralized and solidified. The sludge was transferred to DOT-approved 55 gallon drums for disposal at CECOS/CER near Cincinnati, Ohio. The acid process tanks and storage containers were decontaminated and tested for cyanide and sent to a local salvage yard for reclamation. The third week of the cleanup included the excavation and removal of the contaminated sewer line north of the building. Caustic liquids were removed from process tanks and storage containers from inside the building and transferred to Envirite in Harvey, Illinois. The caustic sludges were neutralized and solidified, then transferred to DOT-approved 55 gallon drums for disposal at CECOS/CER. The caustic process tanks were decontaminated, tested for cyanide-residue and sent to a local salvage yard for reclamation.

The cleanup action concluded with the decontamination of the inside of the building and disposal of solid waste drums at CECOS/CER. Table 1 lists the date and type of material removed from the facility. Total cleanup costs are summarized in Section 1.5. A detailed chronology of daily activities is contained in Appendix U.

1.4 State Efforts to Cleanup Site

Extensive efforts have been undertaken by the MDNR to remedy the groundwater pollution problem because citizens had complained of drinking water problems in October 1978. The MDNR notified Northernnaire's President, Willard S. Garwood that the site was the suspected source of chromium contamination in the two residential wells. He requested the company to inspect all their sewer lines, scrubber vents and begin a hydrogeological investigation. The MDNR sampled soil surrounding the site and requested the facility to develop a PIPP plan. In July 1979, the owner wrote the MDNR stating that Northernnaire did not have funds to proceed with the hydrogeological study. MDNR then initiated enforcement actions for RCRA violations against Acts 245 and 64 which the company had made. The MDNR participated in all site inspections and has the keys to the building. Copies of relevant state correspondence are contained in Appendix V.

TABLE 1
MANIFEST LOG

<u>DATE</u>	<u>AMOUNT</u>	<u>HAULER</u>	<u>DISPOSAL FACILITY</u>	<u>MANIFEST #</u>
7/20/83	Acid liquid N.O.S. 3450 gal.	Nelson Ind.	Nelson Ind.	MI0325954
7/28/83	waste cyanide UN 1935/4000 gal.	Envirite	Envirite	MI 0328354 IL 0820917
7/29/83	waste cyanide UN 1935/1402 gal.	Envirite	Envirite	MI 032853 IL 0816932
8/1/83	waste cyanide mix 66 drums	Inland Waters	CECOS/CER	MI 0328357
8/2/83	waste hypochlorite solution/5000 gal.	Nelson Ind.	Nelson Ind.	MI 0328355
8/3/83	waste cyanide mix 70 drums	Inland Waters	CECOS/CER	MI 0328358
8/3/83	waste cyanide mix 14 drums	Inland Waters	CECOS/CER	MI 0317213

1.5 Cost Summary/Incident Obligation Log

Costs incurred for this Immediate Removal Action totaled \$156,171.58. Expenditures are summarized in the following Tables 1 through 4.

TABLE 1 - Contractor Costs per Petrochem Invoice,
Contract # 68-95-007, Ken Banaszek, OSC.

<u>Item</u>	<u>Amount</u>
Labor	45,489.39
Materials	14,399.30
Equipment	33,432.05
Subcontractor	<u>48,720.28</u>
Subtotal	\$142,041.02

TABLE 2 - Itemization of Petrochem Expenditures
for Subcontractors

<u>Subcontractor</u>	<u>Amount</u>
Rusty Johnson (security)	\$ 3,364.34
Envirite Corporation	13,240.36
Environmental Consultants, Inc.	5,229.87
Smiley's Refuse Service	2,214.23
Frankies Septic Tank	163.80
City of Cadillac	40.20
Nelson Industrial	5,769.08
Leo Ingraham Jr.	1,310.40
CECOS	<u>17,388.00</u>
Subtotal	48,720.28

TABLE 3 - Itemization of EPA and TAT Expenditures

<u>Item</u>	<u>Amount</u>
EPA Labor (Ken Banaszek)	\$ 3,362.44
EPA Travel	2,140.95
EPA Miscellaneous	36.26
TAT Labor and Direct Expenses	8,589.15
Trailer rental TAT Spec. Proj. 5-8307-S1	<u>650.00</u>
Subtotal	\$14,778.80

TABLE 4 - Summary of Overall Expenditures for the
Northenaire Immediate Action.

<u>Item</u>	<u>Amount</u>
Contractor Expenditures	\$142,041.02
EPA and TAT Expenditures	<u>14,780.56</u>
TOTAL	\$156,821.58

1.6 Community Relations

A public meeting was held at the Cadillac City Hall at 1300 hrs., 6 July 1983 (See list of attendees in Appendix T). Site cleanup actions, disposal practices, site safety and contingency planning for emergency evacuation were discussed. The meeting was chaired by Ken Banaszek, OSC, U.S. EPA and Don Renny, Director, Cadillac Utilities. No media representatives attended the initial public meeting. Media coverage began on 11 July 1983 after the cleanup action commenced. The OSC granted periodic interviews to the media throughout the cleanup. The public and local officials appeared satisfied with the emergency action procedures.

1.7 Public Health

The City of Cadillac well field and two contaminated private wells are within one-quarter of a mile of the Northenaire plant site. There was a concern that the city wells could become contaminated. The city wells are 250 to 300 feet deep. The contaminated private wells are shallow wells and are closer to the suspected source of contamination -- the connection of the Northenaire sewer to the city sewer.

In addition to drinking water contamination, there was a risk of public exposure to heavy metals in the waste piles and soils at the site. There was also a potential for leakage of acids and caustics from the building. Chemical burns could have resulted from physical contact with the wastes. Finally, the presence of both acid and cyanide wastes in the building could have resulted in the production of hydrogen cyanide, a highly toxic gas.

The chemical wastes were removed from the building and disposed of and the building was decontaminated. The sewer connection from the facility to the sanitary sewer was removed. The building was then secured with locks to prevent access. Only the contaminated soil still remains at the site and will presumably be addressed in remedial cleanup actions.

2.0 EFFECTIVENESS OF REMOVAL ACTIONS

2.1 Responsible Parties

On 18 October, 1982 the U.S. EPA sent letters to James Zane of Top Locker, Willard S. Garwood, and Robert W. Meyer explaining U.S. EPA plans to initiate field investigations, feasibility studies and removal of hazardous wastes under CERCLA. All parties were offered the opportunity to perform voluntarily the required work and to abate the release of hazardous substances. On 18 November 1982, Willard S. Garwood responded to the 18 October letter, stating Northernnaire was technically bankrupt. On 12 November 1982 James Olson responded to the 18 October letter for R.W. Meyer, Inc., requesting a meeting with EPA. No response was ever received from Top Locker.

On 28 and 29 June 1983, OSC Banaszek contacted potentially responsible parties by telephone and issued verbal notification of EPA's intent to initiate an Immediate Action at Northernnaire. Potentially responsible parties were given 24 hrs. to voluntarily accept responsibility for the cleanup. No potentially responsible parties expressed either a willingness or ability to undertake a cleanup.

2.2 State and Local Officials

The Michigan Department of Natural Resources, Cadillac, office conducted the site investigations that revealed soil and groundwater contamination from the facility. The MDNR instructed Mr. Willard S. Garwood, President of Northernnaire, to bring his facility into compliance with federal and state discharge regulations. The MDNR assisted representatives during several site inspections after the facility was abandoned. Local officials revoked Northernnaire's discharge permit and cut off their sewer connection.

MDNR representatives monitored the Northernnaire cleanup and arranged for disposal of Class III refuse at a local landfill (see Appendix T). Local officials also provided assistance to the U.S. EPA in procuring vermiculite and lime. Local officials helped in identifying utility connections to buildings and assisted in community relations programs.

2.3 Federal Agencies

On 16, and 17 March 1983, a site inspection was conducted by the U.S. EPA, TAT and MDNR. The inspection indicated that the site was not an immediate threat to human health and environment, but rather a potential threat. It was recommended that the liquid, drums and sludge be removed as soon as possible to prevent future discharges from the facility. A 10-point document was prepared for the site. Upon approval, the U.S. EPA was quick in procuring the service of the Petrochem Services, Inc. to remove the liquid drums and sludge from the site. TAT was tasked to assist the OSC with cleanup activities (Appendix F.)

2.4 Contractors

Petrochem Services, Inc., and its subcontractors performed the work required of them in an expedient and cost effective manner. Initiation of the work was timely and the quality of the work was satisfactory. The entire cost of the project was completed below the budget ceiling.

3.0 PROBLEMS ENCOUNTERED

3.1 General

In general, the cleanup proceeded smoothly with only a few minor problems encountered.

3.2 Weather

Severe thunderstorms slowed outside decontamination work on several occasions at the site.

3.3 Sewerline Removal

Inspection of the sewer line revealed that improper connections were made on the sewer line at several locations. Because of these findings, on 27 July 1983 the sewer line was excavated and removed from the site. The connections were photographed and are contained in Appendix P.

Removal of the line revealed the following:

- o A concrete catch basin connection 20 feet north of the building was actually a dry well.
- o The concrete catch basin and two associated metal pipes entering the catch basin were not sealed.

- o The metal pipe exiting northeast from the concrete catch basin entered segmented concrete drain tiles. This junction was not sealed.
- o The concrete drain tiles were fitted end to end but were not sealed in any fashion.
- o Discolored soil was evident surrounding the sewer line.

3.4 Catch Basin Discovered Inside Building

Two catch basins were discovered inside the building after all the tanks were removed. One of these was filled with concrete. The second was connected to the sewer line that exited the north side of the building. Careful inspection of this second catch basin revealed it was of similar construction to the one outside the building, and apparently a dry well. This catch basin dry well probably concentrated soil and groundwater contamination under the building.